

Q1: Integrated Visualization [R]

				Continent
C1	12.5	92	140	Asia
C2	8.2	88	65	Europe
C3	21.0	95	330	N. America
C4	3.1	75	210	S. America
C5	5.6	80	95	Africa

- Questions:
1. Write an R program to create a bubble chart using appropriate aesthetic mappings.
 2. Select suitable aesthetics for GDP, Literacy Rate, Population, and Continent.
 3. Interpret the relationship between GDP and Literacy Rate.
 4. Identify the continuous and categorical variables.
 5. Recommend one improvement to enhance the visualization's readability.

Q3: Nonlinear Axes [R]

	App Downloads
1	50
2	95
3	180
4	400
5	820
6	1600
7	3100

- Questions:
1. Write an R program to display the data using a linear y-axis.
 2. Modify the program to display the same data using a logarithmic y-axis (scale_y_log10()).
 3. Compare both visualizations.
 4. Explain why a logarithmic scale is suitable for this dataset.
 5. Identify one situation where a logarithmic scale should not be used.

Q5: Coordinate Systems (Cartesian and Polar) [R]

Q2: Mapping Data onto Aesthetics [R]

			Category
11	320	4.2	Electronics
12	540	3.8	Grocery
13	210	4.6	Fashion
14	690	4.0	Grocery
15	150	4.8	Fashion
16	410	3.5	Electronics

- Questions:
1. Write an R program (using ggplot2) to create a scatter plot with Orders on the x-axis and Rating on the y-axis.
 2. Map Category using an appropriate aesthetic (e.g. colour) and Orders using another suitable aesthetic (e.g. size).
 3. Interpret the relationship between Orders and Rating.
 4. Identify the type of scale used for each variable (nominal, ordinal, continuous).
 5. Suggest one alternative visualization that better represents the same data.

Q4: Legends and Annotations [R]

			Co. Z (Rs)
Jan	410	380	450
Feb	430	395	460
Mar	460	410	440
Apr	480	430	470
May	520	460	510
Jun	500	490	530

legend. 2. Annotate (using `annotate()`/`geom_text()`) the point of highest price for each company.
3. Explain the role of legends in multi-series visualizations.
4. Discuss best practices for placing legends to avoid obscuring data.
5. Suggest an alternative to a legend for distinguishing categories (e.g., direct labeling).

Questions:

1. Write an R program to plot the monthly stock-price trend for all three companies on a single chart with a clear

	Store Footfall
6 AM	10
9 AM	45
12 PM	80
3 PM	60
6 PM	95
9 PM	70
12 AM	20
3 AM	5

Questions:

1. Write an R program to visualize the dataset using a Cartesian coordinate system (bar chart).
2. Write another R program (using `coord_polar()`) to visualize the same data on a 24-hour clock-style Polar coordinate system.
3. Compare the effectiveness of both visualizations.
4. Explain which coordinate system is more appropriate for cyclical/time-of-day data.
5. Justify your answer based on data interpretation.

Q1: Integrated Visualization [R]

Given Dataset

```
data1 <- data.frame(
  Country = c("C1", "C2", "C3", "C4", "C5"),
  GDP = c(12.5, 8.2, 21.0, 3.1, 5.6),
  Literacy = c(92, 88, 95, 75, 80),
  Population = c(140, 65, 330, 210, 95),
  Continent = c("Asia", "Europe", "N. America", "S. America", "Africa")
)
```

data1

1. Write an R program to create a bubble chart using appropriate aesthetic mappings.

```
library(ggplot2)

ggplot(data1, aes(x = GDP, y = Literacy,
  size = Population, color = Continent)) +
  geom_point(alpha = 0.7) +
  labs(title = "GDP vs Literacy Rate",
    x = "GDP (Rs Tr)",
    y = "Literacy (%)",
```

```
size = "Population",
color = "Continent") +
theme_minimal()
```

2. Select suitable aesthetics for GDP, Literacy, Population and Continent.

- GDP → X-axis position
- Literacy → Y-axis position
- Population → Bubble size
- Continent → Colour

3. Interpret the relationship between GDP and Literacy Rate.

The visualization shows that countries with higher GDP generally have higher literacy rates. For example, C3 has the highest GDP and literacy rate. C4 has the lowest GDP and also a lower literacy rate.

4. Identify the continuous and categorical variables.

Continuous variables:

- GDP
- Literacy
- Population

Categorical variables:

- Country
- Continent

5. Recommend one improvement to enhance visualization's readability.

Use different colours for continents, suitable bubble sizes, clear axis labels, and transparency to avoid overlapping bubbles.

Q2: Mapping Data onto Aesthetics [R]

Given Dataset

```
data2 <- data.frame(
  Item = c("I1","I2","I3","I4","I5","I6"),
  Orders = c(320,540,210,690,150,410),
  Rating = c(4.2,3.8,4.6,4.0,4.8,3.5),
  Category = c("Electronics","Grocery","Fashion",
               "Grocery","Fashion","Electronics")
)
```

data2

1. Write an R program using ggplot2 to create a scatter plot with Orders on x-axis and Rating on y-axis.

```
library(ggplot2)

ggplot(data2, aes(x = Orders, y = Rating)) +
  geom_point(size = 4) +
  labs(title = "Orders vs Rating",
       x = "Orders",
       y = "Rating") +
  theme_minimal()
```

2. Map Category using an appropriate aesthetic and Orders using another suitable aesthetic.

```
ggplot(data2, aes(x = Orders, y = Rating,
                  color = Category, size = Orders)) +
  geom_point(alpha = 0.8) +
  labs(title = "Orders, Rating and Category",
       x = "Orders",
       y = "Rating",
       color = "Category",
       size = "Orders") +
  theme_minimal()
```

- Here:
- **Orders** → X-axis and size
 - **Rating** → Y-axis
 - **Category** → Colour

3. Interpret the relationship between Orders and Rating.

There is **no strong direct relationship** between orders and ratings. Items with high orders do not always have the highest ratings. For example, 14 has the highest orders, while 15 has the highest rating.

4. Identify the type of scale used for each variable.

Variable	Scale
Item	Categorical
Orders	Continuous
Rating	Continuous
Category	Categorical

5. Suggest one alternative visualization that better represents the same data.

A **bar chart** can be used to compare orders across items, with colour representing category.

```
ggplot(data2, aes(x = Item, y = Orders, fill = Category)) +
  geom_col() +
  labs(title = "Orders by Item",
        x = "Item",
        y = "Orders") +
  theme_minimal()
```

Q3: Nonlinear Axes [R]

Given Dataset

```
data3 <- data.frame(
  Week = 1:7,
  App_Downloads = c(50,95,180,400,820,1600,3100)
)

data3
```

1. Write an R program to display the data using a linear y-axis.

```
library(ggplot2)

ggplot(data3, aes(x = Week, y = App_Downloads)) +
  geom_line() +
  geom_point(size = 3) +
  labs(title = "App Downloads by Week",
        x = "Week",
        y = "App Downloads") +
  theme_minimal()
```

2. Modify the same data using a logarithmic y-axis.

```
ggplot(data3, aes(x = Week, y = App_Downloads)) +
  geom_line() +
  geom_point(size = 3) +
  scale_y_log10() +
  labs(title = "App Downloads by Week - Log Scale",
        x = "Week",
        y = "App Downloads (Log Scale)") +
  theme_minimal()
```

3. Compare both visualizations.

- The **linear scale** shows the actual difference in download values.
- The **logarithmic scale** compresses large values.
- The log scale makes the growth pattern easier to observe across all weeks.

- The linear graph makes the later high values dominate the visualization.

4. Explain why a logarithmic scale is suitable for this dataset.

A logarithmic scale is suitable because the number of downloads increases rapidly from **50 to 3100**. The values span a large range, so the log scale makes both small and large values easier to compare.

5. Identify situations where a logarithmic scale should not be used.

A logarithmic scale should generally not be used when:

- Data contain zero or negative values.
- Exact differences between values are important.
- The audience may not understand logarithmic scales.
- The data have only a small range of values.

Q4: Legends and Annotations [R]

Given Dataset

```
data4 <- data.frame(
  Month = c("Jan","Feb","Mar","Apr","May","Jun"),
  Co_X = c(380,430,460,480,520,500),
  Co_Y = c(380,395,410,430,460,490),
  Co_Z = c(450,460,440,470,510,530)
)
```

data4

1. Write an R program to plot monthly stock-price trends for all three companies on a single chart with a clear legend.

```
library(ggplot2)

data_long <- data.frame(
  Month = rep(data4$Month, 3),
  Company = rep(c("Co. X","Co. Y","Co. Z"), each = 6),
  Price = c(data4$Co_X, data4$Co_Y, data4$Co_Z)
)

ggplot(data_long, aes(x = Month, y = Price,
                      group = Company, color = Company)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  labs(title = "Monthly Stock Price Trends",
       x = "Month",
       y = "Stock Price (Rs)",
       color = "Company") +
  theme_minimal()
```

2. Annotate the highest price for each company.

```
ggplot(data_long, aes(x = Month, y = Price,
                      group = Company, color = Company)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  annotate("text", x = 5, y = 520,
    label = "Co. X Max = 520", vjust = -1) +
  annotate("text", x = 6, y = 490,
    label = "Co. Y Max = 490", vjust = -1) +
  annotate("text", x = 6, y = 530,
    label = "Co. Z Max = 530", vjust = -1) +
  labs(title = "Monthly Stock Price Trends",
    x = "Month",
    y = "Stock Price (Rs)",
    color = "Company") +
  theme_minimal()
```

3. Explain the role of legends in multi-series visualizations.

Legends help the viewer **identify and distinguish different data series**. In this graph, the legend identifies Co. X, Co. Y and Co. Z.

4. Discuss best practices for placing legends to avoid obscuring data.

- Place the legend where it does not cover important data.
- Use a clear title for the legend.
- Keep the legend close to the graph.
- Avoid unnecessary legends.
- Use distinct colours or symbols for different series.

5. Suggest an alternative for gendering distinguishing categories.

An alternative is to use **different line types or point shapes**, such as:

```
ggplot(data_long, aes(x = Month, y = Price,
                      group = Company,
                      linetype = Company,
                      shape = Company)) +
  geom_line() +
  geom_point(size = 3) +
  labs(title = "Stock Price Trends",
    x = "Month",
    y = "Stock Price (Rs)") +
  theme_minimal()
```

Q5: Coordinate Systems — Cartesian and Polar [R]

Given Dataset

```
data5 <- data.frame(
  Hour = c("6 AM", "9 AM", "12 PM", "3 PM",
    "6 PM", "9 PM", "12 AM", "3 AM"),
  Footfall = c(10,45,80,60,95,70,20,5)
)
```

1. Write an R program to visualize the dataset using a Cartesian coordinate system (bar chart).

```
library(ggplot2)

ggplot(data5, aes(x = Hour, y = Footfall)) +
  geom_col() +
  labs(title = "Store Footfall by Hour",
       x = "Hour of Day",
       y = "Store Footfall") +
  theme_minimal()
```

2. Write an R program using coord_polar() to visualize the same data on a 24-hour clock-style polar coordinate system.

```
ggplot(data5, aes(x = Hour, y = Footfall)) +
  geom_col() +
  coord_polar() +
  labs(title = "Store Footfall - 24 Hour Clock") +
  theme_minimal()
```

3. Compare the effectiveness of both visualizations.

Cartesian bar chart:

- Easy to read.
- Easy to compare exact footfall values.
- Best for quantitative comparison.

Polar chart:

- Shows the data in a circular 24-hour pattern.
- Useful for identifying busy and quiet periods.
- Exact comparisons are slightly harder.

4. Explain which coordinate system is more appropriate for cyclical/time-of-day data.

The **polar coordinate system** is more appropriate when the main purpose is to show the **cyclical nature of time** because the end of the day connects back to the beginning.

However, the **Cartesian bar chart** is better for accurate comparison of footfall values.

5. Justify your answer based on data interpretation.

The store has the **highest footfall at 6 PM (95)** and the lowest at **3 AM (5)**. The polar chart clearly represents the daily cycle, while the Cartesian chart makes the numerical differences easier to compare.